MULUNGUSHI UNIVERSITY
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MACHINE LEARNING — QUESTIONS AND ANSWERS

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QUESTIONS AND ANSWERS

Question 1: Facebook identifies your friend from an album of photographs (Face Recognition). Which ML class does this fall under?

Answer: Supervised Learning (Classification)

The model is trained on labeled images (photos tagged with names) and learns to classify a face to a known identity.

Question 2: Netflix recommends a movie based on your past choices (Recommendation Systems). Which ML class does this fall under?

Answer: Unsupervised Learning (Clustering / Collaborative Filtering)

The system discovers hidden patterns among users with similar tastes without any labeled data.

Question 3: Analysing bank data for suspicious transactions and flagging fraud (Fraud Detection). Which ML class does this fall under?

Answer: Supervised Learning (Binary Classification)

The model is trained on historically labeled transactions (fraud / not fraud) and learns to classify new transactions accordingly.

Question 4: A system that automatically adjusts MRI scanner parameters to optimize image quality, learning from feedback on resulting images. Which ML class does this fall under?

Answer: Reinforcement Learning

The system acts as an agent that adjusts parameters, receives image quality as a reward signal, and learns to improve through trial and error.

Question 5: An intelligent system that assists radiologists in segmenting brain tumors from MRI scans, learning from the radiologist's corrections. Which ML class does this fall under?

Answer: Reinforcement Learning (with Active Learning elements)

The system adapts based on the radiologist's corrections, which serve as the reward signal guiding improvement over time.

SUMMARY

#	**Scenario**	**ML Class**
1	Facebook Face Recognition	Supervised Learning
2	Netflix Recommendations	Unsupervised Learning
3	Bank Fraud Detection	Supervised Learning
4	MRI Scanner Adjustment	Reinforcement Learning
5	Brain Tumor Segmentation	Reinforcement Learning